Rutgers University
Fall 2021
Math 151 Sections 54-56

Workshop 3

Problem 1
Keywords: Limits, Indeterminate Forms, Cancellation

(a) Evaluate $\lim_{x\to 1} x - 1$ and $\lim_{x\to 1} x^2 - x$.

(b) Can we use the limit laws to evaluate $\lim_{x\to 1} \frac{x^2 - x}{x - 1}$? Explain your answer using 1–2 English sentences.

(c) We do not want to give up on this limit! Factor the numerator. As long as $x \neq 1$, we can cancel a factor. How can we rewrite $\frac{x^2 - x}{x - 1}$ *whenever* $x \neq 1$?

(d) Using your previous answers, explain why $\lim_{x\to 1} \frac{x^2 - x}{x - 1}$ exists and compute its value.

(e) Use the same factoring procedure to try to evaluate the following limits. Note that direct substitution always gives the same 'answer' initially.

$$\lim_{x\to 3} \frac{(x-3)^2}{x^2 - 9} \qquad \lim_{x\to 2} \frac{x^2 - 4}{(x-2)^3} \qquad \lim_{x\to -2} \frac{x^2 + 3x + 2}{x^2 + 4x + 4} \qquad \lim_{x\to 1} \frac{x - 1}{\frac{2}{x} - 2} \qquad \lim_{x\to 3} \frac{\sqrt{x+1} - 2}{x - 3}$$

Problem 2
Keywords: Limit Laws, Indeterminate Forms

If $\lim_{x\to 0} \frac{f(x)}{x^2} = 1$, find the following limits, whenever possible. Don't just guess – give an argument using the limit laws and some cleverness to determine the limits.

(a) $\lim_{x\to 0} f(x)$

(b) $\lim_{x\to 0} \frac{f(x)}{x}$

(c) $\lim_{x\to 0^+} \frac{\sqrt{|f(x)|}}{x}$ (Hint: if $x > 0$, then $x = \sqrt{x^2}$.)

(d) Is f continuous at 0? Write 1-2 English sentences to justify your conclusion.

Problem 3.
Keywords: Limits, Does not Exist, Functions

For each part below, give an example of functions that satisfy the stated conditions and graph them. In each case, the domain of the functions should be *all* real numbers.

(a) $\lim_{x\to 2} f(x) = 3$ and $f(2) = 4$.

(b) $\lim_{x\to -1} h(x)$ exists, and its value is $h(-1) + 2$.

(c) $\lim_{x\to 2} u(x)$ and $\lim_{x\to 2} v(x)$ do not exist, but $\lim_{x\to 2} (u(x) + v(x))$ exists and is equal to 1.